

Research Article

Measuring Social Trust in AI: How Institutions Shape the Usage Intention of AI-Based Technologies

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What drives people to have trust in using artificial intelligence (AI)? How does the institutional environment shape social trust in AI? This study addresses these questions to explain the role of institutions in allowing AI-based technologies to be socially accepted. In this study, social trust in AI is situated in three institutional entities, namely, the government, tech companies, and the scientific community. It is posited that the level of social trust in AI is correlated to the level of trust in these institutions. The stronger the trust in the institutions, the deeper the social trust in the use of AI. To test this hypothesis, we conducted a cross-country survey involving a total of 4037 respondents in Singapore, Taiwan, Japan, and the Republic of Korea (ROK). The results show convincing evidence of how institutions shape social trust in AI and its acceptance. Our empirical findings reveal that trust in institutions is positively associated with trust in AI technologies. Trust in institutions is based on perceived competence, benevolence, and integrity. It can directly affect people's trust in AI technologies. Also, our empirical findings confirm that trust in AI technologies is positively associated with the intention to use these technologies. This means that a higher level of trust in AI technologies leads to a higher level of intention to use these technologies. In conclusion, institutions greatly matter in the construction and production of social trust in AI-based technologies. Trust in AI is not a direct affair between the user and the product, but it is mediated by the whole institutional setting. This has profound implications on the governance of AI in society. By taking into account institutional factors in the planning and implementation of AI regulations, we can be assured that social trust in AI is sufficiently founded.

1. Introduction

One of the most fundamental dimensions in the relationship between human and modern technology is trust. In any field of technology, trust plays a greater role than any other human factors in the usage of technological products and systems [1]. In emerging technologies, it is trust that gives people confidence to fly with airplanes, to live in high-rise buildings, or to ride on high-speed railways [2]. In a broad picture, trust is always present in the utilization of technology, as it mediates the interaction between the user and the

technical object in everyday life [3, 4]. In the field of artificial intelligence (AI), the question about trust matters even more. As the world is seeing a growing application of AI, AI-driven technologies are rapidly changing human life today [5, 6]. A cacophony of digital devices and infrastructures designed with machine learning capabilities gradually penetrates into the most intimate space of human relations and activities, suggesting what music to listen, what food to order, and which persons to be friended [7, 8]. More importantly, AI enters a new phase where our ability to make decisions is being replaced by a set of algorithms,

which allow the machine to work on its own logic, often causing social dilemmas [9].

The present study explores into this area of sociotechnical systems where people are pressed to give “consent” to the logic of AI embedded in technological infrastructures on which their daily life is deeply dependent. We focus our study on AI-driven public services where AI becomes the basis for providing and delivering essential services to people. We are interested in unpacking the relationship between AI and society in three major sectors, namely, transportation, healthcare, and finance. These three sectors are selected due to the nature of the risk involved, which people must consider prior to using it. More specifically, our study is aimed at examining trust in the adoption of AI manifested in three technological innovations, namely, autonomous vehicles (AVs), automated healthcare (AH), and personalized finance technology (fintech). The developments and utilization of AI in these areas are rapidly undergoing. It revolves around the goal of achieving better systems of operation with lesser risk. Thus, AI was adopted into these systems to replace human operators, which are considered to have penchants for making mistakes.

Our study is aimed primarily at measuring the level of trust in AI. Measuring trust in AI is an important task because it can explain to what extent people are willing to rely on AI when making decisions related to their daily activities [10]. Our study contributes to the study of trust in AI by conducting a survey that cuts across three different areas of AI-based technologies. This allows us to compare how trust may differ from one sector of AI to another. A further goal of our study is to examine how trust in AI can serve as a predictor of usage intention of AI [11, 12]. This is another important contribution of our study, as it sheds light on the social factors that determine the level of acceptance of AI in society. As our study is comparative of different AI-based technologies, it helps explain the variety of acceptance in each technology [13].

The most important contribution of our study is about the source of trust in the utilization of AI-based technologies. Using a cross-country survey, our study asks where trust in AI comes from [14]. As shown by Kaplan et al. [15], most studies on trust in AI would consider that trust results from cognitive perception in the performing ability of AI. In these streams of thought, sociopsychological approaches are commonly used to treat trust in AI as being generated primarily from how people think about the technical qualities of the machine [16]. This predisposition draws on the microlevel interaction between the user and AI systems where trust is considered a direct outcome of individual perception [17]. While measuring trust as a cognitive response to AI is acceptable in its context, our study is designed to go beyond the micro realm by bringing it to the institutional contexts where trust is constructed. We argue that trust in AI is shaped by institutions that exist outside the cognitive realm. Further, the way trust in AI is built is strongly influenced by how institutions provide a favorable environment for AI to be socially accepted and adopted [18].

By institutions, we refer to three entities that play central roles in the production and utilization of AI—the AI ecosys-

tem, namely, the government, tech companies, and the scientific community. Later in this study, we shall explain the role and influence of each institutional entity and the variables used to measure the correlation between trust in institutions and trust in AI. This is how we frame the notion of social trust in AI. The following sections will elaborate this framework.

Another substantial feature of our study we should note is a cross-country comparison. It distinguishes our study from one-country surveys commonly conducted by many researchers (see, e.g., [19]). Comparative perspectives make an important aspect due to our focus on the institutional origin of social trust in AI. By treating social trust in AI as being influenced by the institutional environment, it is strongly relevant to examine how AI is socially trusted across sociopolitical contexts with different institutional conditions. Accordingly, we situate our study in East Asia, where institutions vary from one country to another. This region is witnessing unprecedented development and application of AI. It is a group of industrialized economies, including the Republic of Korea (ROK), Japan, Taiwan, and Singapore. It offers the advantage to pursue this comparative study in times when East Asian societies seem to highly embrace digitalization with AI capabilities. Little is known about whether they are aware of the social, political, and cultural implications of the utilization of AI in everyday activities. Thus, the present study is designed to assess the level of awareness and trust among East Asian people in AI-based systems.

2. Literature Review

Trust is conceptually elusive and for good reason. Trust permeates all levels of human existence and comprises different levels of risk [20]. Indeed, because trust is risky, we tend to consider what trust is and when it is warranted [21]. In this section, we begin by briefly exploring philosophical and psychosocial views of trust. We opt to do this because trust is a value-laden and complex phenomenon which demands care from those who intend to study it. After that, we carry our discussion forward to focus on defining social trust and the specific positionalities that trustors take with respect to AI. At the end of this section, we explicate the knowledge gap in trust literature that motivates the present study.

2.1. Philosophical and Psychosocial Trust. Trust is understood as an attitude, while trustworthiness is understood as a property that we ascribe to a subject of trust [20]. Trust, however, is distinct from mere reliance. Here, Baier [22] points out that trusting can be “betrayed, or at least let down, and not just disappointed” (p. 235). So, reliance, on one hand, requires that an agent is dependent on another to achieve some end state. Trust, on the other hand, may be separate from this dependence: It contains an emotive property. For present purposes, we may reasonably consider trust as a combination of mere reliance and an emotive property, undertaken by a trustor from a position of vulnerability toward a trustee (see also [23]). That is, trust arises when

the said emotive property is ascribed in conjunction with reliability to a trustee.

Now, whereas a philosophical understanding of trust undergirds our understanding trust as beyond mere reliance, a social scientific understanding of trust will also assist us in understanding how trusting transactions take place. This is because toward the present purpose, we should not only seek to understand the nature of trust, but also how trust is transacted in our interactions with institutions and technologies. This will serve to provide us with a holistic picture of how we ought to study trust in ways that are most valuable in ameliorating our understanding of our interactions with such nonhuman entities. Psychological approaches to trust investigate trusting behavior in terms of an underlying disposition, intergroup processes, and cognitive expectations [24]. The most common components in psychological trust literature are, similarly to philosophical trust, (1) voluntary risk-taking to make oneself vulnerable to others and (2) expectation of the other's reciprocal response or trustworthiness [25]. In the largest meta-analysis of trust as a psychological construct, Hancock et al. [26] examined trustor factors, trustee factors, and contextual factors. They found that trustors' propensity to trust correlated with the trustor's sense of another individual's trustworthiness (which was mitigated by the trustor's culture and the trustor's performance on the same task that they are entrusting to the trustee). Predictor variables for trustee trustworthiness, unsurprisingly, included trustee expertise, performance, reliability, reputation, and transparency ([26], p. 20). Contextually, the authors found that team communication and perceived trustworthiness were significantly correlated, as were trustworthiness and in-group membership, as well as the presence of shared mental models. Cohesion and conflict were found to also be positively and negatively correlated with trustworthiness, respectively ([26], pp. 16–17). This is similar to a sociological, *relational* perspective of trust, viewing trust “not as a characteristic of an individual, but as an attribute of a social relationship between two or more actors” ([27], p. 191).

Although the construct of trust in the psychosocial literature cited above seems to point to uniquely human factors of trust, our examination into philosophical trust provides us with grounds on which we may position our understanding of trust toward artificial systems as persons—at least in the psychosocial sense. That is, we may at least provisionally contemplate AI as a person in the sense that is relevant for it to be a trustee. This gives rise to several important insights as to the psychological correlative predictors of trust in AI. Applied to the present context, then, trust emerges as a foremost matter of interest in the intra- and interpersonal adoption of AI systems among us. Since trust is context-dependent, significant interest arises relating to how the phenomenon of trust has evolved with technology. Evidence points to an exponential increase in trust as it relates to the employment of e-commerce; more people are using the cyber realm for a greater part of their needs [28].

To take stock of what we have covered so far, then, when we trust an entity, we undertake a position of vulnerability toward a trustee, to whom we also invest an emotively nego-

tiated expectation of reliability. This transaction is influenced by such contextual, psychosocial factors as reputation, transparency, and general reliability. The same, it appears, obtains with technology. Trusting an AV, for example, can be framed as a position undertaken beyond mere reliance. That is, we do not only rely on the AV to get us to our destination; we also rely on a commitment from the AV (or, perhaps more accurately, the manufacturer of the vehicle) to do so safely, comfortably, and otherwise without incident.

As it relates to AI systems, then, the positionalities in which we engage in trusting exchanges with AI are worthy of study. The disruptive potential of AI has been hailed as a potentially fatal, problematic step in human evolution [29] or as the next phase of development in human society with unmatched capabilities for good [30]. This gives rise to a prerogative to understand more deeply our trusting interactions with AI, as trusting AI technology underlies the intention of users for cooperation, economic exchange, and collective action, serving as a foundation for societal and economic development. True to form, Del Giudice et al. [31] have, in a human resource context, also demonstrated that AI acceptance is heavily influenced by trusting attitudes toward technology, such as personality, emotional, and cognitive factors. With this in mind, we now turn to the specific application of social trust in AI.

2.2. Social Trust in AI. The above provides valuable insight into what trust *is*, and the reason for why the low level of public trust in AI systems may hinder receiving the potential benefits of AI. A review by Robles and Mallinson [32] highlighted a gap in the existing frameworks for AI governance—the lack of attention to social trust. The notion of social trust entails a broader context for individuals to build confidence in the AI ecosystem, a framework that encompasses relevant institutions in the construction of trust [33]. Here, trust results from experiencing circumstances that constantly change [34]. So trust is not a fixed condition but rather a product of our response to the social environment. This is the distinctive feature of social trust that we argue has been hitherto understudied. It is an area of importance because it indicates the trustworthiness of the social environment in which we live (e.g., [35]). It is informed by the notion that social trust reflects the conditions of society and social systems [36] in which we are exposed to the application of AI. From this, we see a crucial intersectionality: Trust in AI is not a standalone, direct attitude that individuals hold toward AI systems in a sociological vacuum. Instead, this trust is heavily influenced by sociological and environmental systems across different layers of our social lives. Keeping in mind our exposition on philosophical and psychosocial trust above, we begin to see that social trust in AI is not just negotiated by mere reliance on emotive content; it is subject to shifting sociopolitical systems that influence our trusting relationships with AI.

Indeed, social trust has different levels of operation. At the most macro analysis, social trust may be measured to represent the characteristics of the whole country where nation-states become the unit of analysis (e.g., see [37], p.

99). This is the level where trust is linked to national conditions such as democracy, equality, cohesion, and happiness [38, 39]. It is also possible to construct a fragmentation at the mesolevel where social trust is described in correspondence to the institutional conditions that determines the penetration of AI into social life. The mesolevel of trust is particularly interesting, given its immediate influence on trust-based transactions. Institutional conditions at the mesolevel, as Criado et al. [40] argue, require further examination on the “unique evolution of public organizations, services, and policies” (p. 182). Of note, the authors point out that new patterns and emerging modes of services that are based on automated decision-making across different sectors require further study (in the same place). A small Norwegian study [41] has broken new ground in this topic, finding that among their 20 participants, trust in governmental institutions is indicative of positive attitudes toward AI systems. A conceptual study on the mediative effects of trust in AI [42] has also argued for the influential power that digital communications—mitigated by public policy—have on trust and, therefore, AI adoption.

This, together with the understandings of philosophical and psychosocial trust that we have discussed earlier, motivates the kind of social trust that we wish to propound in this study. We define social trust in AI as a positional attitude of reliance that a trustor holds toward an AI—the *trustee*—which includes an undertaking of risks pertaining to vulnerability and emotive investment, mitigated or otherwise affected by larger institutional contexts. This is in line with a study by Wu et al. [43] that refers social trust to “the willingness to rely on those [institutions] who have responsibility for making decisions and taking actions related to the managing and regulating the focal issues.” The importance of institutions in the construction of trust in AI cannot be overstated. Modern society revolves around institutions which contribute to the establishment of order and production of wealth. How societies are able to make progress depends on the quality and capacity of institutions to serve their functions for the people (e.g., see [44]). The character of large, political economic institutions transform the action of individuals into collective actions in pursuit of the common goal of society [45]. Every institutional entity has a role to play and contributes to the production of a specific area in social life. Some of them hold authority across sectors due to its centrality in the social structures (in the same place). According to a global study conducted by the University of Melbourne and KPMG [46], “institutional pathway” that entails the importance of governance, regulation, and public confidence in institutions’ ability to safeguard against potential risks associated with AI is underlying for social trust in AI.

The intertwined nature of social and technical elements complicates the definition of social trust in AI. Following the sociotechnical framework suggested by Geels [47], we focus on elements that define the landscape of technology production. Thus, we position social trust in AI in constant evolution together with policy development, market logics, and innovation trajectory. Policy development defines how the government views the significance and meaning of AI,

which informs policies on AI development and adoption. Market logics relate to how business organizations consider AI as a key instrument to enhance their profitability through products and services they offer to users in the market. The innovation trajectory represents the direction of research and development that the scientific community takes to make progress in the field of AI. Trust in the whole institution that creates and operates the AI ecosystem is fundamental to social acceptance of AI. This is a kind of trust individuals and groups decide to give to social entities whose access and authority have an impact on users. In this domain, three social entities will be tested and measured: the government agency responsible for regulating AI, business organizations that provide AI-based services and products, and the scientific community that designs the AI ecosystem. However, in this study, we do not investigate the actors involved in the application side of AI technology, for example, media, societal groups, and users, and their impact on social trust in AI.

To take stock of our present discourse on trust in AI, then trust is a known mediator of our interactions with each other. Such interactions can be understood via myriad pathways, among which is a layered view of social trust. To this end, recent literature gives rise to reasons to examine trust on a macro- and mesolevel. How do institutions mediate our trusting interactions with AI systems? To understand our transactions with AI technology, we propose in this study that we should examine trust in ways that are as robust as the cited literature shows. We argue that trust is reflected in human assessment of the dedication of the institution to serving the public interest generally, considering the respondents’ experiences utilizing the institution’s goods and services (public interest) [48], the amount of resources used to achieve the institutional goal (effectiveness) [48], and the ability to fulfill its vision and goals (capability) in a transparent way (honesty) in compliance with stated goals and objectives (sincerity) [49]. However, trust in AI technologies requires confidence not only in the intentions and competencies of institutions but also in the transparency, reliability, and ethical soundness of the technology itself. To this end, then, we argue that we should examine our trust in the ability of AI technologies to treat all users equally and in a fair manner (fairness), increase the comprehensiveness of functioning and decision-making (transparency), deliver consistent and uninterrupted service (robustness and resilience), and ensure transparent data collection, protection, and utilization (data monetization, regulation, and storage) (Afroogh, 2024). The role and contribution of each institutional entity shall be elaborated in the following section.

To sum, in this section, we have made clear our motivation to examine social trust and its relationship with AI technologies as it is mitigated by social institutions. There is a significant body of literature available on social trust relating to institutions and AI; further study is warranted on (1) the extent to which the different components of social trust interact with each other and (2) the extent to which different types of institutions are accorded social trust and how different variables of trust may predict or influence such distributions. To do this, inspired by our examinations of

philosophical and psychosocial trust, we take social trust as our subject of interest, specifically referring to the willingness to rely on commitments that AI systems undertake toward their users within the broader institutional and societal framework, acknowledging the role of government, tech companies, and the scientific community in shaping trust.

3. Conceptual Framework and Hypotheses

By incorporating social trust as a central pillar in the adoption and application of AI, we developed a framework to test several hypotheses. It seeks to fill a lacuna in understanding how the aggregated influence of trust in institutions shapes social trust in AI-based technologies. The core framework is illustrated in Figure 1. The framework is constituted by three areas of phenomenon this study has focused on. It starts with *trust in institutions* as an independent variable. The other two are treated as dependent variables, namely, *trust in AI* and *usage intention of AI*. Each of these areas is further elaborated below.

3.1. Trust in Institutions

3.1.1. The Government. The government is an institution responsible for regulating AI through establishing a policy regime. In this study, confidence in government competence, benevolence, and integrity defines trust in the institution of government. Also, it emphasizes on transparency of government actions and robustness of regulatory frameworks as the predictors of trust in government. In the context of social trust in AI, trust in government translates into how competent the government is in producing policies and regulations to maintain user privacy, ensure data protection, and implement ethical guidelines [49].

3.1.2. Tech Companies. Tech companies are considered the institution that produces AI-based services and products guided by market logic. The willingness of the public to engage with AI technologies is heavily influenced by their perception of the tech companies behind these innovations and their contribution to the development of reliable emergent technologies [48]. In this study, trust in tech companies depends on companies' reputation, transparency in operations, competency, and adherence to ethical practices [50, 51]. However, being profit-driven entities, tech companies are often questioned about the clash with societal values, raising concerns about the alignment of developed technology with the public good and benevolence. Accordingly, the components of trust in tech companies in our study focused on their integrity, competence, and benevolence.

3.1.3. Scientific Community. Scientific community is the institution that has developed the foundation of AI. It is also the one that contributes to producing fundamental innovations for the AI ecosystem, creating the innovation trajectory of AI-based technologies. The credibility of the scientific community plays a crucial role in establishing social trust in AI technologies. Its ability to challenge its own work and produce transparent and objective research results strengthens its trustworthiness [52]. The commitment of

the scientific community to transparency through peer review and replication of results amplifies the level of trust [53]. Scholars agree on expertise (competence), integrity, and benevolence as the main contributors to defining public trust in scientists [54].

Interdependencies of the aforementioned institutions are crucial in recognizing the complex sociotechnical landscape in which AI is situated and highlight the role of institutions in accelerating a trust-infused environment conducive to the adoption of AI technologies.

3.1.4. Variables. To measure the level of trust in each of the institutional entities described above in a coherent fashion, we developed five variables as the matrix to be included in the questionnaire. Each variable is elaborated as follows:

- a. *Public interest* (Tx01) refers to the commitment of the institution to serving the interest of the public in general given the experience of respondents in using the products and services provided by the institution. Specifically for the government, it relates to the use of its power to help those in need; for tech companies, it is about priorities on the interest of their consumers; and for the scientific community, it touches on how their works make people's lives better.
- b. *Effectiveness* (Tx02) refers to the extent to which resources are mobilized to achieve the goal of an institution. The more effective an institution, the fewer resources are spent, and the greater the impact it makes in society. Specifically for the government, it relates to the reporting system on the results of its programs; for tech companies, it is indicated by products with a positive impact on society; and for the scientific community, it deals with scientific breakthroughs with high-quality innovation.
- c. *Capability* (Tx03) refers to the ability of the institution in carrying out its mission and in implementing its vision for the purpose of serving the public. This ability is reflected by the real impact felt by the respondent. Specifically for the government, it is measured by the quality of public services delivered; for tech companies, it is embodied in comprehensive attention to detail to ensure public safety; for the scientific community, it is the ability to find scientific solutions to social problems.
- d. *Sincerity* (Tx04) refers to the level of achievements in compliance with stated objectives. It asks questions about the process and procedures each institution has gone through to deliver public needs. Specifically for the government, it is reflected in the results that fulfill outlined promises; for tech companies, it relates to the ways they deliver what they promise and keep their commitment; and for the scientific community, it focuses on sharing their discoveries for the betterment of society.
- e. *Honesty* (Tx05) refers to the level of openness and transparency in carrying out its duty and fulfilling

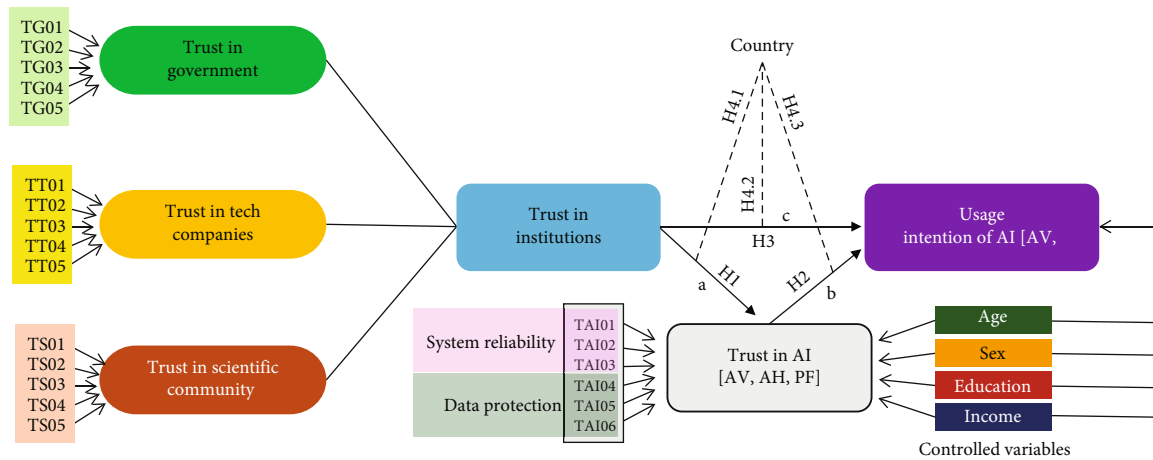


FIGURE 1: Conceptual model showing the direct effect of trust in institutions and trust in AI on usage intention of AI and the mediating effect of trust in AI on usage intention of AI. The dashed arrow shows the moderating effect of countries.

its societal goals. This aspect covers areas of information and communication with the public and how their internal processes are communicated sufficiently to the public. Specifically, for the government, it opens updates and information on administrative actions; for tech companies, it deals with the disclosure of information in order to inform the public extensively; and for the scientific community, it shares with the public any scientific development as well as mistakes and problems.

H1: Trust in institutions is positively associated with trust in AI technologies (i.e., AV, AH, and personalized finTech [PF]).

3.2. Trust in AI-Based Technologies. AI is often viewed as a paradigm-shifting technology able to enhance human capabilities at low cost [55]. Human productivity, effectiveness, and business processes are augmented by the emergent feature of AI that underlies the development of other technologies [56]. Advancements in AI-assisted healthcare [57], transportation [58], education [59], and commerce [60] revamp the everyday experiences of users. However, the issue of trust in technology and thus its adoption remains persistent due to its debatable nature. For example, Gille et al. [61] highlight the overspilling feature of trust; as if citizens do not trust the government to protect their data, they will not trust the healthcare system for the same. To gauge a better understanding of trust constructs, the authors suggest breaking it down to variables such as transparency, awareness, and reproducibility of clinical trials with AI-driven technologies. Trust in AVs is suggested to be measured through the interaction of dispositional trust (user's tendency to trust automation in general), situational trust (trust shown by a user of an automated system, in a context-specific situation), and dynamic learned trust (trust developed during system use, based on user's skills and knowledge) [62]. A similar approach is employed by Zarifis and Cheng [63], where the correlation among an individual's psychological disposition to trust, sociological factors

influencing the individual's trust, trust in financial organizations, and trust in AI is used to measure trust in finTech.

3.2.1. AI-Based Technologies. In this study, we have selected three AI-based technologies that represent the nature of trust in relation to institutional environments. As we know, the exposure of individuals to AI-driven technologies is growing rapidly due to the broad penetration of personalized recommendations (e.g., Amazon and Netflix), chatbots or virtual assistants, self-driving vehicles, wearable devices (e.g., smartwatches and fitness trackers), robo-advisors, microfinance apps, etc. into everyday lives. To capture the understanding of social trust in AI, our survey is designed to measure the nuances in perceptions toward three AI-driven technologies (AVs, AH, and PF) that are in the process of swift digitalization in the selected countries. The choice of these domains is determined by their current and potential impact on society by offering solutions with lesser risk. AVs have the perspective to revolutionize transportation [64], making it safer and more reliable [65, 66]. AH promises improved diagnostic capabilities and personalized treatment options [67–69], undermining data privacy and the safety of automatic decisions [70]. PF services offer opportunities for a range of services: from individual wealth management to credit risk assessments [71, 72], requiring adequate personal data protection and transparency [73].

3.2.1.1. AV. The introduction of AVs on public roads pertains to the enhancements of human life, mobility, sustainability, and inclusivity. With the majority of road accidents happening due to human errors, switching to AVs has the potential to increase road safety [74]. AVs can improve the quality of life for elderly and disabled individuals by providing more mobility options. They can also impact human productivity due to being freed from driving time. Optimized route planning performed by AVs can reduce travel times and contribute to a more efficient traffic flow; therefore, leading to a decrease in carbon footprint [75]. At the same time, concerns about data protection, safety, and security persist for connected car systems [76]. Moreover,

unresolved ethical dilemmas that impose decision-making power on algorithms call for more transparency.

3.2.1.2. AH. The exponential increase in medical data collected from wearable devices and by healthcare professionals has reformed medical experiences. The integration of AI and healthcare advances the development of new diagnostic and prognostic tools [77]. The vast amount of patient data collected aids in the development of individualized treatment techniques, which improve therapy outcomes. Patients' economic and health outcomes can be improved by providing them with more tailored services [78]. Big data repositories and image recognition have seen significant use in medical sectors such as pathology [79], cancer imaging [68], and nuclear medicine [69]. Healthcare practitioners use AI-driven predictive analytics to detect and avert health issues, which reduce inpatient costs.

3.2.1.3. PF. The utilization of finTech products has boosted recently, triggered by enhanced connectivity among every person, organization, product, service, and activity anywhere, at any time, through various channels (e.g., QR codes, Wi-Fi networks, mobile applications, social media networks, and intelligent digital assistants). Personalized financial tools include automated, secure, risk-averse, and fragility-averse banking businesses and services (e.g., mobile banking), tailored insurance products, lending, loan and mortgage services, accessible online markets and trading, and smart internet finance (e.g., digital currencies, P2P lending, and crowdfunding) [72]. These AI-driven systems offer intelligent review, planning, generation, and recommendations of financial objectives and actions. They enable anytime access to heterogeneous networks and applications, suggest novel opportunities such as new products, functions, and markets, and understand user intention, preference, and feedback.

3.2.2. System Reliability. System reliability refers to the experience of faultless operation of AI systems, being tied to how safe AI-powered technologies are. We explore trust in AI system reliability by looking at how willing users are to believe that using AI carries minimal risk of breakdown or danger [80]. Despite the perceived superiority of AVs over human drivers that is attributed to dynamic adaptation to external conditions, technological risks brought along should be mitigated through ensuring safety, liability, privacy, and cybersecurity [81]. In the analysis of AI incident cases reported on the website AI Incident Database, found that more than 60% of all incidents are attributed to the AI reliability events, with more than 35% of them resulting in deaths or injuries. Kyriakidis et al. [64] highlight the problem of software hacking and manipulation of vehicles by adverse actors as an underlying contributor to public concerns regarding the use of AVs. To quantify AI system reliability in AVs, disengagement events (when the AI system or backup driver determines that the human driver needs to take over control) serve as essential metrics [81]. It was also highlighted that training AVs in various environments ensures vehicle safety and system robustness and increases user adoption [82]. The rapid integration of AI in finTech platforms has triggered new challenges, particularly in ensuring the integrity, confidentiality, privacy,

and availability of data, that can undermine the social trust in AI-driven technology [72]. More transparency in algorithms used in finTech was established to correlate with increased trust and reputation gains for companies [83]. Unintentional data bias, often rooted in racial, gender, and age bias, and insufficient model validation also pose operational and reputational risks, resulting in declined trust in technology [84]. A significant impediment to the broader adoption of AI systems in healthcare settings and cautious acceptance by patients is often referred to as the lack of interpretability [85]. According to Breiman [86], the ultimate goal should be obtaining accurate and reproducible results to expedite providing medical treatment. Assurance of AI's fairness and unbiased performance becomes important, and verifying data validity and analyzing model performance on subgroups are essential tools to address the concerns [87] of accuracy and reproducibility.

3.2.3. Data Protection. The second variable of trust in AI-based technologies is data protection. It ensures user acceptance of AI in the domains of transportation, healthcare, and finances. It mandates the establishment of robust data privacy protocols, authentication processes, and authorization mechanisms that govern data collection, retention, processing, and transmission. McDermott [88] defines privacy, autonomy, and nondiscrimination as the main values that underpin data protection as a fundamental right. Unauthorized practices involving the collection, utilization, access, storage, and sharing of information pose a substantial threat to personal data integrity, compromising user privacy [89]. If users are confident in proper data protection, they are more likely to engage with the respective service [90], boosting innovation in these areas. High-profile data breaches (e.g., Cambridge Analytica that misused Facebook data to influence the elections) seed apprehension toward data protection by tech companies and governments and destroy public trust in the service. Informing users in a transparent and straightforward way about how data is collected, stored, protected, and used expedites adopting AI-driven technologies. Moreover, data protection by design indicates that safeguarding users' rights does not impede technological advancement.

3.2.4. Variables. Based on two key aspects of trust in AI explained above, we formulated six variables for each aspect to measure the level of trust in the AI-based technologies examined in this study.

- a. Fairness (TAI01) refers to how AI-based technologies are designed to treat all users fairly and equitably, regardless of their demographic and socioeconomic background. Specifically, to support equal access to PF, financial systems are trained to avert discrimination and provide equal access to loans, insurance, banking, and other financial services. Likewise, to guarantee that all patients receive appropriate care, treatment, and therapies, AH should be free from biases in diagnosis to impede the aggravation of existing social inequalities. AI algorithms of AVs should be trained to distinguish all road users and treat them fairly to maintain traffic safety.

- b. Transparency (TAI02) refers to how AI-based technologies contribute to strengthening users' understanding of how these systems operate, make decisions, and use data. Equipping users with this knowledge facilitates making informed judgments about how they interact with these technologies. Transparency in pricing, loan and insurance structuring, risk assessments, and algorithmic trading helps prevent hidden costs and eliminate unfair practices, strengthening confidence in financial digital platforms. Transparency about using data collected by wearable devices and healthcare practitioners and its contribution to AI-assisted diagnoses and therapy recommendations raises patients' awareness about their health conditions. Patients can actively engage in decision-making regarding their own health being able to derive the data-driven rationale. Transparency about decision-making processes and safety features deployed in AVs helps alleviate public fears about handing over control to AI-driven technology. Importantly, the ease of auditing and verifying transparent AI systems by independent parties gauges additional accountability to them.
 - c. Robustness and resilience (TAI03) refer to system robustness as another key indicator of system reliability. During periods of market instability or high-volume trading, robust financial service exhibits consistent performance and maintains uninterrupted transaction processing. Robust AI-driven healthcare ensures that diagnostic tools and patient management systems remain consistent, reproducible, and reliable across a range of clinical scenarios and external interventions. Employed by AVs, robust AI algorithms assure safe operations across diverse driving circumstances and unpredictable situations. Moreover, robust systems withstand cybersecurity threats and data misuse. They quickly recover after system failures, retaining their initial effectiveness that drives higher confidence in using these services.
 - d. Data monetization (TAI04) refers to users' data protection and the transparency of the intentions to use the collected data, which has become a cornerstone of public trust because AVs are designed to exchange information both with sensors along the route (external sharing) and with the manufacturer (internal sharing) to ensure safe driving. FinTech organizations handle substantial amounts of personally identifiable information. However, oftentimes, data can be misused and sold to third parties, offering economic benefits to the companies, leading to a decrease in users' trust in AVs and fintech. The prospect of monetization of healthcare data, while holding potential for medical advancement, raises fears among users about the potential compromise of medical confidentiality and privacy.
 - e. Data regulation (TAI05) refers to establishing clear guidelines for data collection and usage, ensuring that companies in these sectors handle sensitive information responsibly. Data regulation often mandates security standards and accountability measures, reducing the risk of data breaches and cyberattacks that can erode public trust. In healthcare, regulations ensure patient confidentiality, which is essential for maintaining trust in medical institutions and encouraging honest communication between patients and healthcare providers. For AVs, data regulations help establish standards for collecting and sharing vehicle performance data, which is crucial for improving safety and gaining public acceptance. In fintech, regulations promote transparency and give consumers more control over their financial data, fostering trust in digital financial services.
 - f. Data storage (TAI06) refers to methods used to collect and protect user data. In fintech, secure data storage safeguards sensitive financial information, protecting consumer privacy and combating fraud. For healthcare, reliable data storage solutions protect patient records, allowing for accurate diagnoses and treatment while ensuring anonymity. AVs, in turn, record and analyze massive volumes of sensor data critical for refining safety algorithms and investigating incidents. It requires cloud processing that impacts user privacy.
- 3.3. *Intention to Use AI.* Conceptually, trust is inexplicably linked to usage rates and willingness to engage with technological systems. A proper trust relationship has been identified as a requirement for effective AI-assisted decision-making, most notably in healthcare [91]. Accordingly, a lack of trust in AI systems comprises a significant drawback in the adoption of this technology, especially in risk-involving sectors such as healthcare, transportation, and finance. It should be noted that this may be influenced by several human factors such as user education, past experiences, user biases, and perceptions toward automation. AI system properties such as controllability, transparency, and complexity of the model, associated risks, and the like also influence trust in AI uptake [91]. Hengstler et al. [92] found similar results in medical and transportation AI, highlighting the symbiosis of trust in the technology as well as in the innovating firm and its communication about the technology as a mediator of trust in AI systems. All these provide a solid basis upon which we may surmise that social trust is strongly associated with rates of AI deployment and overall digitalization in society.
- H2: Trust in AI-based technology (AV, AH, and PF) is positively associated with the intention to use AI technologies.
- H3: Trust in institutions is positively and directly associated with the intention to use AI technologies (AV, AH, and PF) and indirectly via higher trust in AI technologies.

In addition to three hypotheses elaborated above, our survey is complemented by the following hypothesis with three subhypotheses.

H4: The relationships between trust in institutions and trust in AI technologies and user intention on AI technologies will be moderated by countries.

H4.1: The relationship between trust in institutions and trust in AI technologies (Path a) will be moderated by countries.

H4.2: The relationship between trust in AI technologies and user intention on AI technologies (Path b) will be moderated by countries.

H4.3: The relationship between trust in institutions and user intention on AI technologies (Path c) will be moderated by countries.

4. Methodology

4.1. Data Collection. Data for this study were collected by using a questionnaire survey from four East Asian countries: Singapore, Taiwan, the ROK, and Japan, utilizing stratified random sampling to reflect each country's demographic composition in terms of age, gender, education, and ethnicity (for Singapore only) so that it can be representative at the national level. The study received ethical approval from the Institutional Review Board (IRB) at Nanyang Technological University, Singapore (IRB Reference Number: IRB-2023-359). The survey commenced in the first week of November 2023 and concluded on 12 December 2023. Participants were provided with an information sheet outlining the objectives and procedures of the study, followed by an informed consent form, which they were required to sign before participation. The survey instrument, initially developed in English by adopting the existing measures that had been validated in previous studies, was translated into Japanese, Korean, and Taiwanese to ensure cross-cultural equivalence. All items in the survey questionnaire were rated on a 5-point Likert, Likert-type, or semantic differential scale.

4.2. Participants. Singapore, the ROK, Japan, and Taiwan have been using digital technologies such as AI, the Internet of Things (IoT), and robotics to address growing urban challenges and promote social, economic, and environmental sustainability [93]. All four places apply AI to the transportation, healthcare, and financial sectors. According to the Government AI Readiness Index reports, Singapore, the ROK, Japan, and Taiwan are East Asian regional leaders in implementing and governing AI technologies ([94–97]). Singapore, the ROK, and Japan, respectively, ranked first, second, and third in East Asia in the Government AI Readiness Index report for 4 consecutive years (2021–2024), while Taiwan ranked fourth in 2022 and 2024 and fifth in 2021 and 2023 ([94, 95]; Hankins et al. 2023; [97]). Assessing the level of awareness and trust among people in these four places can provide useful references for countries or cities that are keen on implementing AI in the transportation, healthcare, and financial sectors.

The sample size for each country is as follows: 1017 (Japan), 1012 (the ROK), 1005 (Singapore), and 1003 (Tai-

wan), totaling 4037 participants for the entire study. Specifically, there were 1973 men (49.02%) and 2052 women (50.98%). The participants were from Age 21 and above, with a mean age of 45 years. In terms of education levels, 269 participants (6.66%) had no formal education or less than primary-level schooling, while 867 (21.47%) completed primary education. A total of 573 participants (14.19%) attained secondary education, and 523 (12.95%) completed postsecondary or tertiary education. The majority, 1511 participants (37.42%), held a bachelor's degree or equivalent, followed by 244 (6.04%) with a master's degree or equivalent, and 50 participants (1.23%) who obtained a doctoral degree or equivalent. In terms of household income, 339 respondents (8.40%) reported earning \$1000 or less per month, while 757 participants (18.75%) had a monthly income between \$1001 and \$3000. A larger group of 1013 individuals (25.09%) fell within the \$3001–\$5000 range, and 717 respondents (17.76%) earned between \$5001 and \$8000. Additionally, 530 participants (13.13%) reported incomes between \$8001 and \$10,000, and 681 (16.87%) had monthly incomes exceeding \$10,000.

4.3. Instruments. The theoretical constructs in the study were measured using multiple items rated on a 5-point Likert scale, where higher values indicated more positive responses. To confirm the reliability of these measurements, Cronbach's alpha coefficients were computed for each scale. The specifics of how each variable was derived and the corresponding items are detailed in the sections that follow.

Trust in institutions was assessed across three domains: trust in government, trust in tech companies, and trust in scientific communities, with each domain comprising five items on a 5-point Likert scale to capture various aspects of trust:

Trust in government: Trust in government (mean = 2.98; SD = 0.44) is measured by five items, which include perceptions of public interest (TG01), effectiveness (TG02), capability (TG03), sincerity (TG04), and honesty (TG05). The internal reliability is satisfactory (Cronbach's $\alpha = 0.94$).

Trust in tech companies: Trust in tech companies (mean = 3.30; SD = 0.46) is measured by five items: public interest (TT01), effectiveness (TT02), capability (TT03), sincerity (TT04), and honesty (TT05). The internal reliability is satisfactory (Cronbach's $\alpha = 0.89$).

Trust in scientific communities: Trust in scientific communities (mean = 3.47; SD = 0.44) is measured by five items measured on the public interest (TS01), effectiveness (TS02), capability (TS03), sincerity (TS04), and honesty of scientists and researchers (TS05). The internal reliability is satisfactory (Cronbach's $\alpha = 0.84$).

Trust in AVs: Trust in AV (mean = 3.27; SD = 0.54) is measured by six items which included operational independence (TAI01AV), security from unauthorized access (TAI02AV), adaptability to environments (TAI03AV), data minimization (TAI04AV), data sharing for safety (TAI05AV), and user access to data (TAI06AV). The internal reliability is satisfactory (Cronbach's $\alpha = 0.84$).

Trust in AH: Trust in AH (mean = 3.33 ; SD = 0.50) is measured by six items, which include the alignment with human doctors (TAI01AH), consistency across providers (TAI02AH), fairness (TAI03AH), data privacy (TAI04AH), data deletion compliance (TAI05AH), and adherence to data protection regulations (TAI06AH). The internal reliability is satisfactory (Cronbach's $\alpha = 0.85$).

Trust in PF: Trust in PF (mean = 3.33 ; SD = 0.43) is measured by six items, which included effective financial management (TAI01PF), security during unauthorized access (TAI02PF), transparency in emergencies (TAI03PF), compliance with data protection laws (TAI04PF), privacy of user data (TAI05PF), and data security (TAI06PF). The internal reliability is satisfactory (Cronbach's $\alpha = 0.90$).

Control variables: Control variables included age collected for individuals aged 21 and above and treated as a continuous variable; gender is treated as dummy coded with 0 for male and 1 for female. Education levels were assessed through a detailed scale ranging from no formal education to doctoral or equivalent, across a 10-point scale. Household income was categorized into six ranges from \$1000 and below to over \$10,001, adjusted for local currencies (SGD, TWD, JPY, and KRW) to maintain consistency across Singapore, Taiwan, Japan, and the ROK. Descriptive statistics, presented in Table 1, report *mean trust levels* across countries for each type of institution (e.g., government, tech companies, scientific community) and each application of AI (AVs, AH, and PF).

4.4. Analytical Approach. The theoretical model shown in Figure 1 suggests that trust in institutions directly predicts trust in AI (Path a) and user intention to use AI (Path c) and indirectly via trust in AI (a*b). Trust in AI is hypothesized to mediate the effect of trust in institutions on usage intention. Additionally, the model indicates that these structural linkages can be moderated by country. Given the multifaceted nature of the constructs, the study employs latent constructs to capture their complexity rather than combining dimensions or creating composite scores. Structural equation modeling (SEM) is widely accepted for such analyses. The hypothesized model includes a moderating variable (country), so we conducted a multigroup analysis in SEM to determine whether differences in structural relationships among countries are due to actual differences and not measurement errors [98]. To evaluate the model fit, we report the chi-square statistic and other fit indices, such as the comparative fit index (CFI), root mean squared error of approximation (RMSEA), and standardized root mean square residual (SRMR). While chi-square is a traditional measure for assessing model fit, its sensitivity to sample size is well documented. Acceptable model fit is indicated by $CFI \geq 0.90$ and RMSEA and SRMR values < 0.08 .

4.4.1. Measurement Model. We conducted a confirmatory factor analysis (CFA) to evaluate the measurement properties of the scales used, based on the entire sample. CFA confirms the hypothesized measurement model, using multiple indicators for each latent construct to reduce measurement error. Our model included seven latent constructs, each

defined by multiple indicators, allowing latent variables to correlate freely. Parameters and fit indices were estimated using the maximum likelihood method:

$$x_1, x_2, \dots, x_n = \lambda_x TI + \epsilon_x,$$

$$y_1, y_2, \dots, y_n = \lambda_y TA + \epsilon_y,$$

$$z_1, z_2, \dots, z_n = \lambda_z IU + \epsilon_z.$$

Here, (x, y) and (z) are observed indicators, (λ) s are factors loadings, and (ϵ) s are errors for the trust in institutions, trust in AI, and intention to use AI, respectively. TI is the trust in institution, TA is the trust in AI technologies, and IU is the intention to use AI.

4.4.2. Structural Model. After validating our measurement model, we estimated the structural model depicted in Figure 1 to test hypotheses H1, H2, and H3. Each latent factor was scaled by fixing one indicator per factor to 1.0, a standard SEM practice for model identifiability. Using the maximum likelihood estimation method, we calculated all parameters and evaluated the model fit:

$$TA = \beta_1 TI + \xi_1,$$

$$IU = \beta_2 TA + \beta_3 TI + \xi_2.$$

Here, (β_1, β_2) and (β_3) are path coefficients that need to be estimated, and (ξ) s are residual errors in the equations.

4.4.3. Multigroup Analysis. Hypotheses H4, H4.1, H4.2, and H4.3 proposed that relationships between trust in institutions, trust in AI technologies, and user intentions regarding AI will be moderated by the national context. To verify country-specific differences, we assessed measurement invariance across countries to ensure the uniform functioning of measurement constructs. This involved using multigroup analysis with chi-square difference tests across a sequence of nested models, starting with a configural model, which estimates parameters separately for each group without imposing equality constraints. This baseline model serves as the reference for successive models that incrementally introduce constraints to test for types of invariances. If the chi-square difference test ($\Delta\chi^2$) is not significant, it indicates structural equivalence between groups. Given the sensitivity of $\Delta\chi^2$ to sample size and distributional properties, we also employed the CFI difference as an alternative measure of invariance. Following [99], a change in CFI less than or equal to 0.01 between models suggests negligible differences in fit, supporting structural equivalence across groups.

Once measurement invariance is established, the moderating effect of country is tested through a multigroup SEM approach. Moderating effects are examined by constraining structural path coefficients to be equal across groups (i.e., countries). If the constrained model (where path coefficients are forced to be equal across groups) significantly differs from the unconstrained model, it indicates the presence of a moderating effect. Initially, a baseline model of best fit is established for each group separately, with all factor loadings

TABLE 1: Descriptive statistics showing the mean trust scores (1–5 Likert scale) across various institutions and AI domains, disaggregated by country.

	Japan (<i>N</i> = 1017)	South Korea (ROK) (<i>N</i> = 1012)	Singapore (<i>N</i> = 1005)	Taiwan (<i>N</i> = 1003)	Overall (<i>N</i> = 4037)
Trust in government	2.5	2.59	3.7	3.13	2.98
Trust in tech companies	3.21	3.18	3.44	3.37	3.3
Trust in scientific communities	3.39	3.38	3.58	3.55	3.47
Trust in autonomous vehicles (AVs)	3.17	3.15	3.46	3.3	3.27
Trust in automated healthcare (AH)	3.17	3.21	3.5	3.43	3.33
Trust in personalized fintech (PF)	3.2	3.23	3.43	3.47	3.33

Note: Full item wording, individual item means, standard deviations, unstandardized factor loadings, and Cronbach's α for each construct are available in Table S1.

specified as invariant across groups. This baseline model is then compared to a fully constrained model where all hypothesized structural paths (e.g., trust in institutions to trust in AI, trust in AI to user intention in AI, and trust in institutions to user intention in AI) are constrained to be equal across groups. A significant chi-square difference between these models suggests differences in path coefficients between groups, indicating a moderating effect of the country. To identify specific paths contributing to observed group differences, the fully restricted model is compared with a series of less restrictive models, where one specific path at a time is allowed to vary freely across groups. Chi-square differences between the fully constrained model and each less restrictive model help determine which paths significantly contribute to observed group differences.

$$TA_c = \beta_{1c} TI_c + \xi_{1c},$$

$$IU_c = \beta_{2c} TA_c + \beta_{3c} TI_c + \xi_{2c}$$

where (*c*) indicates the country group and (β_{1c} , β_{2c}) and (β_{3c}) are path coefficients significantly different across countries.

5. Results

5.1. Measurement Model. A CFA on the entire sample showed a good fit, with the following fit indices: robust CFI = 0.973, robust TLI = 0.971, robust RMSEA = 0.033, and robust SRMR = 0.025. The chi-square statistic was significant ($\chi^2 = 2974.629$, *df* = 486, $p < 0.001$), expected due to the large sample size (Table 1).

5.2. Overall Structural Model and Hypothesis Testing. The SEM results for the proposed path model show a good fit across all three AI domains, as summarized in Figures 2, 3, and 4. The fit indices for each domain are within acceptable ranges, indicating strong empirical support for the theoretical model (Figures 2, 3, and 4). The first assumption of this model is that people's trust in institutions such as governments, scientific communities, and technology companies influences how much they trust AI-based technologies

(AV, AH, and PF), as proposed in H1. This assumption is strongly supported across all three models: The path from institutional trust to domain-specific trust in AI (labeled Path *a*) is statistically significant and substantial in size (e.g., 0.52 for AV, 0.59 for AH, and 0.63 for PF). This supports the notion that widespread trust in institutions is a key foundation for trust in AI technologies.

The model further demonstrates that trust in each specific AI technology is a strong predictor of intention to use it, as predicted in H2. The Path *b* coefficients across all domains are statistically significant and sizeable (e.g., 0.86 for AV, 0.67 for AH, and 0.69 for PF). In other words, when people trust an AI system's capability in a specific context such as driving a car, managing their health, or handling their finances, they are more likely to intend to use it.

The third hypothesis (H3) suggests that institutional trust may influence usage intention not only indirectly through domain-specific AI trust but also directly. H3 is partially supported. In the case of AH, institutional trust shows a small (0.11) but statistically significant direct effect on usage intention. However, the direct effect is insignificant and near zero for AVs, indicating that institutional trust influences usage intention primarily through trust in AVs. Similarly, for PF, the direct effect of institutional trust on usage intention is small and insignificant, indicating that institutional trust largely influences usage intention through trust in PF. The subsequent mediation analysis underscores the central role of domain-specific trust in influencing the usage intention of AI. For AV, 96% of the effect of institutional trust on usage intention is indirect, indicating near-complete mediation. This means that institutional trust shapes usage intention almost entirely through trust in AVs. For AH and PF, the indirect effects remain dominant (80% and 76%, respectively), but a meaningful proportion of the effect is also direct, suggesting that institutional trust continues to influence usage intention in domains involving personal well-being.

5.3. Multigroup Analysis of Structural Models and Hypothesis Testing. To assess whether country moderates the relationships in our model or the hypothesized country differences, it is essential to establish that variations among countries

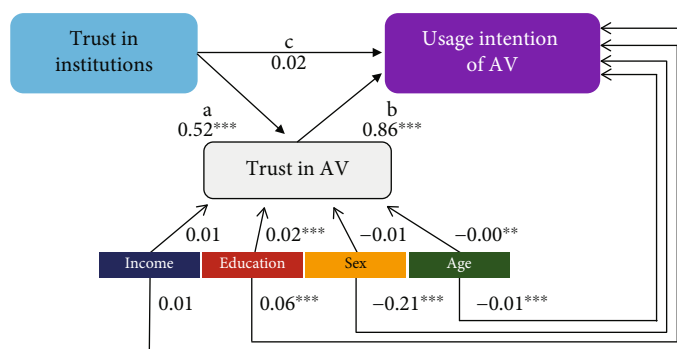


FIGURE 2: Unstandardized path coefficients showing direct, indirect ($a * b = 0.44^{***}$), and total ($[a * b] + c = 0.46^{***}$) effects of the hypothesized structural model for autonomous vehicles (AVs) for the full sample, $N = 4025$. $*p < 0.05$; $**p < 0.01$; $***p < 0.001$. Fit indices: $\chi^2 = 2022.32$, $df = 284$, $p < 0.001$; CFI = 0.968; TLI = 0.964; RMSEA = 0.039; SRMR = 0.037.

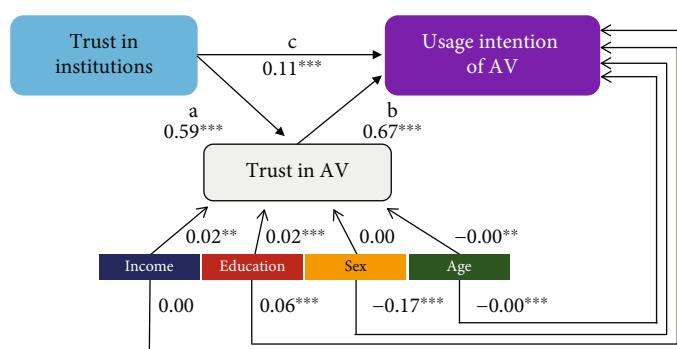


FIGURE 3: Unstandardized path coefficients showing direct, indirect ($a * b = 0.40^{***}$), and total ($[a * b] + c = 0.50^{***}$) effects of the hypothesized structural model for automated healthcare (AH) for the full sample, $N = 4025$. $*p < 0.05$; $**p < 0.01$; $***p < 0.001$. Fit indices: $\chi^2 = 2194.93$, $df = 284$, $p < 0.001$; CFI = 0.951; TLI = 0.945; RMSEA = 0.048; SRMR = 0.040.

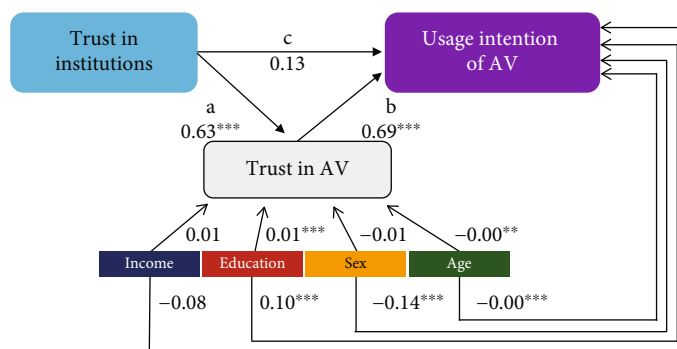


FIGURE 4: Unstandardized path coefficients showing direct, indirect ($a * b = 0.43^{***}$), and total ($[a * b] + c = 0.46^{***}$) effects of the hypothesized structural model for personalized fintech (PF) for the full sample, $N = 4025$. $*p < 0.05$; $**p < 0.01$; $***p < 0.001$. Fit indices: $\chi^2 = 2118.21$, $df = 284$, $p < 0.001$; CFI = 0.969; TLI = 0.965; RMSEA = 0.040; SRMR = 0.037.

exist and that these differences arise from structural variations in the path coefficients, rather than from measurement differences. Accordingly, we conducted measurement invariance analyses using a series of nested models to determine whether the measurement parameters, particularly factor loadings, functioned consistently across countries (see Table S2). The baseline model tested for configural invariance, where model parameters were estimated separately for each group without imposing equality constraints. The configural model provided an adequate fit,

suggesting that the factor structure is consistent across groups. Weak invariance, where factor loadings are constrained to be equal across countries, is supported, as indicated by negligible differences in fit ($\Delta CFI = 0.00$). Strong invariance, where both factor loadings and intercepts are constrained to be equal across groups, is partially supported, with minimal differences ($\Delta CFI = 0.01$) within acceptable thresholds. Strict invariance, where factor loadings, intercepts, and means are constrained to be equal across groups, does not hold, as indicated by significant

differences in fit ($\Delta\text{CFI} = 0.02$). These results suggest that measurement invariance exists up to strong invariance, but strict invariance does not hold.

After confirming measurement invariance (Table S2), we estimated structural models for each country to test the moderating effect of country.

For AVs, trust in institutions positively predicts trust in AV, which in turn predicts usage intention across all countries, though trust in institutions does not directly predict user intention. The model fit indices ($\text{CFI} = 0.959$, $\text{TLI} = 0.954$, $\text{RMSEA} = 0.043$, and $\text{SRMR} = 0.038$) indicate a good fit (not reported in the paper). The baseline model is then compared with a fully constrained model, and the chi-square difference test ($\Delta\chi^2 = 11.4$, $\Delta\text{df} = 9$, $p = 0.25$) shows no significant moderating effect of country, suggesting consistent relationships across countries (Table 2). Therefore, H4 is not supported for AV.

For AH, trust in institutions positively predicts trust in AH, which predicts user intention across all countries. Trust in institutions directly predicts user intention only for Japan and the ROK. The model fit indices ($\text{CFI} = 0.951$, $\text{TLI} = 0.945$, $\text{RMSEA} = 0.048$, and $\text{SRMR} = 0.040$) also indicate a good fit (not reported in the paper). The chi-square difference test ($\Delta\chi^2 = 37.613$, $\Delta\text{df} = 9$, $p < 0.0001$) between baseline unconstrained model with fully constrained model indicates a significant moderating effect of country, showing that these relationships vary significantly across countries. This supports H4. Specifically, paths from trust in institutions to trust in AH (H4.1 supported) and from trust in AH to usage intention differ across countries (H4.2 supported), but not the path from trust in institutions to usage intention (H4.3 not supported) (Table 3).

For PF, trust in institutions positively predicts trust in PF, which predicts user intention across all countries. Trust in institutions directly predicts user intention in Japan, the ROK, and Singapore. The model fit indices ($\text{CFI} = 0.963$, $\text{TLI} = 0.96$, $\text{RMSEA} = 0.043$, and $\text{SRMR} = 0.04$) suggest a good fit (not reported in the paper). The chi-square difference test ($\Delta\chi^2 = 30.651$, $\Delta\text{df} = 9$, $p < 0.001$) between the baseline unconstrained model and the fully constrained model indicates a significant moderating effect of country, showing that relationships vary across countries. Therefore, it supports H4. The path from trust in institutions to trust in PF differs across countries (H4.1 supported), but the paths from trust in PF to usage intention and from trust in institutions to usage intention do not vary across countries (H4.2 and H4.3 not supported) (Table 4).

6. Discussion

To the best of our knowledge, this is the first study examining social trust in three types of AI technologies, namely, AV, AH, and PF, in four East Asian countries: Singapore, Japan, the ROK, and Taiwan. Our empirical findings reveal that trust in institutions is positively associated with trust in AI technologies (H1). This means that a higher level of trust in institutions leads to a higher level of trust in AI technologies. Trust in institutions plays a crucial role in building trust in AI technologies. It can directly affect people's trust in

AI technologies. Our study finds that trust in institutions is based on perceived competence (i.e., effectiveness and capacity), perceived benevolence (i.e., serve public interest), and perceived integrity (i.e., sincerity and honesty). We suggest governments, tech companies, and scientific communities carry out the following practices to increase people's trust in them, which in turn increases people's trust in AI technologies. In terms of competence, trust in AI technologies will increase if institutions have the expertise, knowledge, and skills needed to design, develop, and implement AI technologies effectively. In terms of benevolence, trust in AI technologies will increase if institutions can develop and implement AI technologies to solve economic, social, and environmental challenges; improve the well-being of the general public; prevent, tackle, and mitigate harms brought by AI technologies; and use AI technologies in a responsible and ethical manner. In terms of integrity, trust in AI technologies will increase if institutions can be transparent about their use of AI technologies and the benefits and risks of using AI technologies, sincerely use AI technologies to deliver services they have promised, follow and promote ethical principles of AI, and comply with applicable AI-related policies and legislations.

Besides, our empirical findings confirm that trust in AI technologies (AV, AH, and PF) is positively associated with the intention to use these technologies (H2). This indicates that a higher level of trust in AI technologies leads to a higher level of intention to use these technologies. Trust in AI technologies depends on the extent to which people trust that AI technologies can ensure users' safety, security, privacy, and data protection. To increase social trust in AI technologies, we suggest governments, tech companies, and scientific communities prevent and address any safety risks (e.g., AV accidents and medical misdiagnosis) by conducting safety assessment of AI technologies through a series of testing and evaluation. Only AI technologies that pass the testing and evaluation can be adopted by institutions. Security is a core requirement throughout the life cycle of AI systems ([100], p. 5). Governments, tech companies, and scientific communities should adopt a "secure by default" approach to ensure that there are secure design, development, deployment, and operation of AI systems ([100], p. 5) and "implement security controls and mitigations where possible within their models" ([100], p. 7) to address any security risks (e.g., remote hijacking of AVs and cyberattacks). Besides, they should protect personal data and ensure privacy by practicing data minimization and anonymization when training AI systems [101, 102]. They should collect, use, share, and delete data for AI systems in ways that are consistent with national legislation ([103], p. 21).

Our study finds that trust in institutions only has a positive, direct impact on the intention to use AH, but not AV and PF (H3). This implies that trust in institutions is not sufficient to directly and positively affect the intention to use AV and PF. Some studies find that people's intention to use AVs are affected by several factors. A recent study finds that primary factors affecting people's intention to use AV include "social information, perceived ease of use, perceived usefulness, and perceived risk" ([104], p. e0298348).

TABLE 2: Multigroup analysis: Testing for path coefficient invariance across countries for Autonomous Vehicle (AV).

Model comparison	DF	χ^2	$\Delta\chi^2$	RMSEA	Δ DF	p value (χ^2 test)	Moderating role of country
Model 1: Baseline model (factor loadings invariant)	1136	3297.1					
Model 2: Factor loadings and all path coefficients invariant (fully constrained)	1145	3308.5	11.4	0.02	9	0.25	Not found (H4)

TABLE 3: Multigroup analysis: Testing for path coefficient invariance across countries for Automated Healthcare (AH).

Model comparison	DF	χ^2	$\Delta\chi^2$	RMSEA	Δ DF	p value (χ^2 test)	Moderating role of country
Model 1: Baseline model (factor loadings invariant)	1136	3739.2					NA
Model 2: Factor loadings and all path coefficients invariant	1145	3776.8	37.61	0.05	9	0.00	Yes (H4)
Model 3 path coefficient (a): Trust in institution $\rightarrow$ trust in AH (unconstrained)	1142	3764.4	12.42	0.05	3	0	Yes (H4.1)
Model 3 path coefficient (b): Trust in AH $\rightarrow$ usage intention to AH (unconstrained)	1142	3761	15.79	0.06	3	0	Yes (H4.2)
Model 4 path coefficient (c): Trust in institution $\rightarrow$ usage intention to AH (unconstrained)	1142	3773.4	3.372	0.01	3	0.33	No (H4.3)

TABLE 4: Multigroup analysis: Testing for path coefficient invariance across countries for Personalized Fintech (PF).

Model comparison	DF	χ^2	$\Delta\chi^2$	RMSEA	Δ DF	p value (χ^2 test)	Moderating role of country
Model 1: Baseline model (factor loadings invariant)	1136	3236.1					
Model 2: Factor loadings and all path coefficients invariant	1145	3266.7	30.65	0.05	9	0.00	Yes (H4)
Model 3 path coefficient (a): Trust in institution $\rightarrow$ trust in PF (unconstrained)	1142	3250.1	16.69	0.06	3	0.000	Yes (H4.1)
Model 4 path coefficient (b): Trust in PF $\rightarrow$ usage intention PF (unconstrained)	1142	3264.9	1.87	0	3	0.6	No (H4.2)
Model 5 path coefficient (c): Trust in institution $\rightarrow$ usage intention to PF (unconstrained)	1142	3263.1	3.68	0.01	3	0.3	No (H4.3)

Another study finds that perceived enjoyment, perceived trust, and perceived value are three psychological factors directly and positively affecting people's intention to use AVs [105]. Similarly, some studies find that factors which positively affect people's intention to use PFs include perceived benefits [106–108], social norms [106], and perceived trust in fintech [107]. Existing studies indicate that personal factors rather than institutional factors play a more important role in affecting people's intention to use AV and PF. Higher levels of trust in institutions will positively predict the intention to use these technologies (AV, AH, and PF), indirectly via higher trust in AI technologies.

Our study shows that higher trust in institutions indirectly predicts usage intention of all three technologies (i.e., AV, AH, and PF) via higher trust in AI technologies. This indicates that trust in institutions serves as a foundation for developing trust in AI technologies, which in turn drives usage intentions. Since system reliability and data protection constitute trust in AI technologies, we suggest the government, tech companies, and the scientific community enhance the reliability of AV, AH, and PF when being used and safeguard data collected by AV, AH, and PF so as to

increase social trust in AI technologies, which in turn leads to an increase in usage intention of AI technologies.

A comparison between unconstrained and constrained models shows that the moderating effect of a country is significant when it comes to the relationship between trust in institutions and trust in AH, the relationship between trust in AH and intention to use AH, and the relationship between trust in institutions and trust in PF. It indicates that these relationships vary across countries. Besides, our study finds that trust in institutions directly affects people's intention to use AH in Japan and the ROK. We suggest further studies to identify and examine country-specific factors affecting these relationships in Singapore, Japan, the ROK, and Taiwan. Such country-specific factors can be government strategies, operating environment, infrastructures, talent, commercial efforts, research, and development [109]. A study identified technological implementation, policy setting, and economic and medical impact measurement as three success factors of AI implementation in healthcare [110]. Future studies can examine whether these three factors affect trust in AH or intention to use AH in Singapore, Japan, the ROK, and Taiwan. Regarding trust in PF, it can be

affected by regulations or the government's efforts to promote fintech development. In Singapore, for example, FinTech Regulatory Sandbox Guidelines were issued in November 2016 to provide guidance to financial institutions and fintech companies to carry out more fintech experimentation with the requisite regulatory support [111]. In Taiwan, the year of 2015 was declared by the Finance Supervisory Commission as the Year of Fintech to promote fintech development in the country [112], and a fintech regulatory sandbox was introduced in December 2017 [113]. In Japan and the ROK, a regulatory sandbox was, respectively, introduced in 2018 and 2019 to encourage financial innovation [114, 115]. Future studies can examine whether the launch of a fintech regulatory sandbox can positively affect people's trust in PF.

Drawing on the results of our study, we highlight some theoretical and practical implications. As previously noted, this study provides a sociotechnical perspective on AI trust. Our study confirms that people's acceptance of AI technology is influenced by a variety of institutional and systemic elements in addition to personal experience and technical aspects. Drawing on institution-based trust theory [4] and the sociotechnical framework by Geels [47], we show that trust in AI is shaped by public confidence in the institutions that design, govern, and promote these technologies. These include governments, scientific communities, and technology companies. Public perceptions of reliability, data protection, and ethical behavior are closely tied to how these institutions behave. Trust in AI grows when people believe that the entire system behind the technology is competent, transparent, and aligned with the public interest. Specifically, we demonstrate that the perceived competence, integrity, and benevolence of institutions shape trust in AI, which then mediates the effect on intention to use. This emphasis is necessary because AI systems are often powered by opaque, evolving algorithms that are difficult to interpret, even for experts [116]. Most users cannot fully understand how these systems work, how decisions are made, or what risks may be involved. In such situations, trust in institutions becomes a critical factor in technology adoption. People will be more receptive to the risks that come with using AI if they believe that governments, researchers, and companies are operating ethically, transparently, and in the public good. Institutional trust fills the gap left by limited user understanding, encouraging adoption even when individuals cannot directly assess or control the technology. This is particularly relevant in high-stakes domains such as healthcare, transportation, and finance, where algorithmic decisions can have significant personal consequences.

Our results also advance the sociotechnical literature by demonstrating how domain-specific risks influence the public's trust in AI. AI-based healthcare raises ethical and diagnostic concerns; fintech raises issues related to privacy and data protection; AVs are linked to safety and liability risks. These domain-specific perceptions shape how trust is formed via the intersection between social and technical qualities of AI systems.

We also find that context matters. In Japan and the ROK, institutional trust directly shapes behavioral intention,

especially in healthcare. In contrast, trust in institutions plays a more indirect role in fintech and AV adoption, where technical risk or data misuse is more salient. This suggests that cultural values and historical trust in public institutions may influence how institutional trust affects technology adoption. Countries with stronger regulatory systems, government support for AI, and public belief in state capacity to manage AI risks are more likely to show a stronger institutional trust–AI trust linkage.

Three significant policy implications result from these findings. First, trust in AI cannot rely solely on technical reliability; it must be supported by strong institutionalized governance. The regulation of AI systems, risk management, and ethical considerations should all be spelled out in detail in policy frameworks. Governments and associated organizations must clearly and consistently explain these duties to establish and maintain public trust. Second, trust-building policies must be tailored to the specific risks of each AI domain. Healthcare, finance, and mobility involve different public concerns. A single governance model cannot address all these differences. Each sector needs targeted regulations, risk management strategies, and ethical safeguards that reflect its unique challenges. Third, since institutional trust varies by nation, a successful AI strategy must be based on the national context. Policy can concentrate on promoting innovation and expanding AI uses in settings where public institutions are highly trusted. In contrast, when people have less trust in institutions, governments should focus on being transparent, involving the public, and making decisions openly. Trust in AI can only grow if people first trust the institutions responsible for it.

7. Conclusion and Future Work

Like any other technology, AI-based technologies are always socially situated. The level of trust in using these emerging technologies cannot be separated and detached from the social reality in which they operate and serve the people. At the core of their operation is social trust, which fuels people's confidence in utilizing AI-based technologies for their personal and collective purposes. Whether it is for transportation, healthcare, or financial service, our study has confirmed that trust in AI cannot be attributed only to the technical qualities of these engineered systems. While we recognize the role of technical performance in building users' confidence in adopting AI-based technologies, it is extremely important to highlight the fact that the relationships between AI-based technologies and the people are invariably mediated by the institutional environment where these relationships are built and enhanced.

Having said this, we would like to highlight two critical points in regard to our study on social trust in AI. First, institutions do matter in the construction and production of social trust in AI-based technologies. It is necessary to reiterate this notion because there is a strong penchant among policymakers, government officials, and the media to uphold the conviction that AI is such a superior technology that its existence undermines the presence of human organization. Our study in three AI-based technologies provides firm evidence that

trust in AI is not a direct affair between the user and the product, but it is mediated by the whole institutional setting as represented by three entities, namely, the government, tech companies, and the scientific community. This has profound implications for the governance of AI in society. By taking into account institutional factors in the design and implementation of AI regulations, we can be assured that social trust in AI is sufficiently founded.

Second, it is important to realize the complexity in the use of AI-based technologies. Intention to use AI in public services such as transportation, healthcare, and finance varies due to different degrees of perceived risk in each of these sectors. Our study has shown that correlation is established between the nature of risk in these sectors and how the institution in charge of these services is able to provide reliable systems. In this light, diversity appears across countries where these services are available with different qualities. Once again, this confirms the role and impact of the institutional environment in shaping the level of social trust in AI as well as the intention to use AI-based technologies.

To conclude this paper, it is necessary to acknowledge the limitation of our study. The chosen methodology and the selected study countries are not without downsides because they do not capture the complex aspects of social trust in AI, which in itself is a very complex technology. Hence, there are three areas of investigation that we suggest for further studies on social trust in AI. First, it is very important to complement our quantitative survey with qualitative exploration of social trust in AI. We acknowledge that the nature of social trust is deeply entangled with personal experience and can be very sensitive. There are aspects that cannot be quantified by survey questionnaires. Thus, qualitative studies of this subject are equally meaningful and strategic to unpack the pattern of trust and its construction within the institutional environment. Another future study is also significant on how social trust is shaped by the experience of using non-AI services. This line of inquiry has the potential to add to our findings because it allows us to correlate, for instance, how trust in human medical services affects trust in using AI-based medical healthcare. Finally, AI is now becoming a global phenomenon. The ongoing penetration of this technology is unprecedented, and the impact it makes varies from one country to another. Thus, it is empirically intriguing to extend our survey beyond the East Asian region. A comparative study between regions, for instance, East Asia and the Middle East or Western Europe, would be compelling to conduct to learn how the institutional conditions at the regional level may affect the willingness to adopt AI-based technologies.

Data Availability Statement

The data for this article are publicly available under 10.21979/N9/OQF6TW.

Ethics Statement

This study was approved by the Institutional Review Board of Nanyang Technological University (NTU-IRB) in September 2023 with Reference Number IRB-2023-359.

Conflicts of Interest

The authors declare no conflicts of interest.

Author Contributions

Conceptualization: S.A. S.L.; Study design: S.A., S.L., I.T., M.T.; Data curation and formal analysis: S.S.; Writing—original draft preparation: S.A., S.L., S.S., I.T., M.T.; Writing—review and editing: S.A., S.L., S.S., I.T., M.T.; Supervision and funding acquisition: S.A.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. (*Supporting Information*) Results of cross-country surveys.

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